Ans. to the que. no. 1

(a)

The origin of my position is $(0,2)$ $0\hat{i}+2\hat{j}$, which is 2m above from the ground. Now, my friend is on the roof of a building, 7 meters above and 1 meter away from the edge of the building. The horizontal distance from me to the edge of the building is 5 meters. So, my friends position is $6\hat{i}+7\hat{j}$. Now, the position vector me and friend is $(6-0)\hat{i}+(7-2)\hat{j} = 6\hat{i}+5\hat{j}$ Ans.

(b)

Given that,

$v_0 = 15\,ms^{-1}$

horizontal distance to friend, $x = 6m$

vertical distance to friend, $y = 7m$

the balls horizontal distance, $y_0 = 2m$

Now, ~~we know that,~~

$$x = v_0 \cos\theta_0 t$$

$$t = \frac{x}{v_0 \cos\theta_0}$$

$$t = \frac{6}{15\cos\theta_0}$$

Again,

$$y = y_0 + v_0 \sin\theta_0 t - \frac{1}{2}gt^2$$

$$2 = 2 + 15\sin\theta_0 \left(\frac{2}{5\cos\theta_0}\right) - \frac{1}{2} \times 9.8 \times \left(\frac{2}{5\cos\theta_0}\right)^2$$

$$5 = 6\tan\theta_0 - \frac{19.6}{25}\sec^2\theta_0$$

$$5 = 6\tan\theta_0 - \frac{19.6}{25}\left(1 + \tan^2\theta_0\right)$$

$$5 = 6\tan\theta_0 - 0.784 - 0.784\tan^2\theta_0$$

$$0.784\tan^2\theta_0 - 6\tan\theta_0 + 5.784 = 0$$

Now,

$$\tan\theta_0 = \frac{-(-6) \pm \sqrt{(-6)^2 - (4 \times 0.784 \times 5.784)}}{2 \times 0.784}$$

$$= \frac{6 \pm \sqrt{17.86}}{1.568}$$

Now, (+ve)

$\tan\theta_0 = \tan^{-1}(6.52)$

$= 81.28°$

(−ve),

$\theta_0 = \tan^{-1}(1.13)$

$= 48.27°$

Since, we need the larger angle cause we need the ball to be falling when it hit the friend.

Ans. (81.28°)

Given that,

$V_0 = 15 ms^{-1}$

$y_0 = 2m$

$\theta_0 = 81.28°$

$t = 0.2s$

Now,

$$x(t) = x_0 + v_0 \cos\theta_0 t$$

$$= v_0 \cos\theta_0 t$$

$$= 15 \cos(81.28^\circ) \times 0.2$$

$$= 0.455 \text{ m}$$

$$y(t) = y_0 + v_0 \sin\theta_0 t - \frac{1}{2} g t^2$$

$$= 2 + 15 \sin(81.28^\circ) \times 0.2 - \frac{1}{2} \times 9.8 \times (0.2)^2$$

$$= 4.77 \text{ m}$$

(b)

-for velocity,

$v_x = u\cos\theta_0$

$= 15\cos(81.28^\circ)$

$= 2.274\ ms^{-1}$

$v_y = v_0 \sin\theta_0 - gt$

$= 15\sin(81.28^\circ) - (9.8 \times 0.2)$

$= 12.87\ ms^{-1}$

So, the position of the ball at 0.2 seconds

(0.46m, 4.77m)

and the velocity of the ball at 0.2 seconds

($2.274\ ms^{-1}$, $12.87\ ms^{-1}$)

Ans.

(d)

Given that,

$v_0 = 15ms^{-1}$

$\theta_0 = 75^\circ$

$y_0 = 2m$

$y = 7m$

Now,

$$y = y_0 + v_0 \sin\theta_0 t - \frac{1}{2}gt^2$$

$$7 = 2 + 15\sin(75^\circ)t - \frac{1}{2}\times 9.8\times t^2$$

$$4.9t^2 - 14.49t + 5 = 0$$

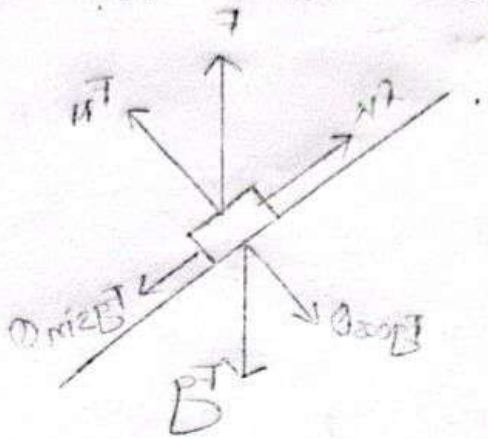
$$t = \frac{-(-14.49) \pm \sqrt{(-14.49)^2 - (4\times 4.9\times 5)}}{2\times 4.9}$$

$$= \frac{14.49 \pm \sqrt{111.928}}{9.8}$$

$$t_1 = \frac{14.99 + \sqrt{111.928}}{9.8} = 2.558\,s$$

$$t_2 = \frac{14.99 - \sqrt{111.928}}{9.8} = 0.398\,s$$

$\therefore t_1$ is supposed to hit him from above.



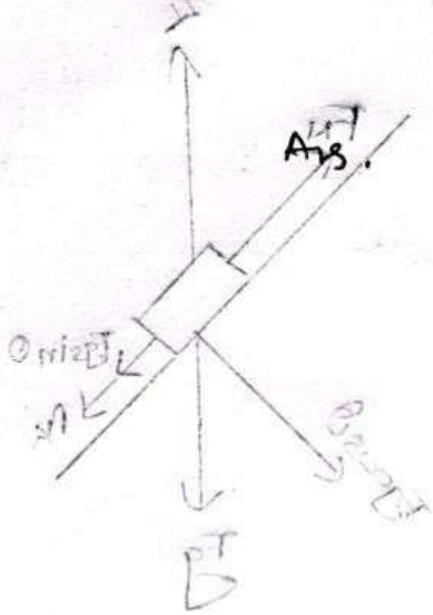
$$x = v_0 \cos\theta_0 t$$

$$= 15\cos(25^\circ) \times 2.56$$

$$= 9.94\,m$$

So, my friend is moved and distance of $x - 6 = 9.9386 - 6 = 3.9386\,m$ and he moved this distance in 2 seconds, his speed was

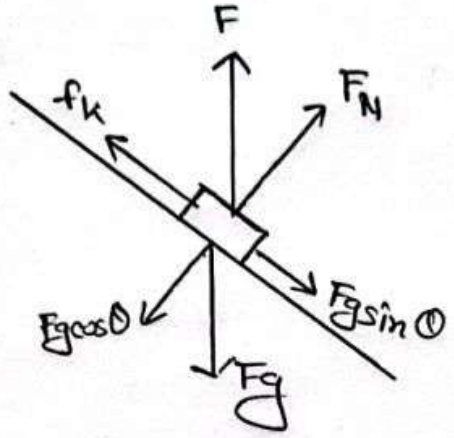
$$\frac{3.94}{2} = 1.97\,ms^{-1}$$



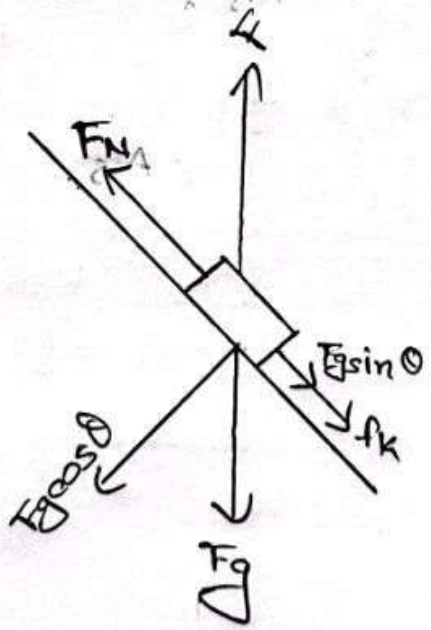
Ans. to the que. no. 2

(a)

for the block going down,



for the block going up,



(b)

Given that,

$m = 5\,kg$

$\theta = 30°$

$w = mg$

$= 5 \times 9.8$

$= 49.05\,N$

$\therefore W_A = W\cos\theta$

$= 49.05 \times \cos(30°)$

$= 42.48\,N$

W_A = the perpendicular of gravitational force

Ans.

Given that,

$m = 5\ kg$

$\theta = 30°$

$F = 100N$

$\mu_k = 0.3$

$\mu_s = 0.5$

$W_B = W\sin\theta$

$= mg\sin\theta$

$= 49.05\ \sin(30)$

$= 24.525\ N$

W_B = Gravitational force component parallel to the plan.

$f_s = \mu_s N$

$= 0.5 \times 42.48N$

$= 21.24N$

$-f_k = \mu_k N$

$= 0.3 \times 42.48$

$= 12.744 N$

To know whether the block moves up or down, we need to compare the forces,

$F_{net} = F - W_B - f_k$

At rest,

$F - W_B = 100 - 24.525$

$= 75.475 N$

$F - W_B > f_s$

So the block will move up (plane).

$F_{net} = 100 - 24.525 - 12.74$

$= 62.73 N$

Now, applying newtons 2nd Law,

$F_{net} = ma$

$a = \frac{62.73}{5}$

$= 12.55 \ ms^{-2}$

Ans